MIL-R-3074D
28 September 1979
SUPERSEDING
MIL-R-3074C
24 February 1972

MILITARY SPECIFICATION

# RAILWAY CAR, GONDOLA: HIGH SIDE 56-1/2-INCH GAGE, 50-TON, 8-WHEEL (WELDED), DOMESTIC SERVICE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a railway high-side gondola car having a welded steel superstructure and a welded steel underframe for domestic service.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-P-3320 | - Painting: Freight and Maintenance Cars. |
| MIL-P-3558 | - Plates, Identification: Locomotives, Railway Cars, Work Equipment, and Maintnenance Cars. |

FSC 2220

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-H28 | - Screw-Thread Standards for Federal Services. |

(Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASSOCIATION OF AMERICAN RAILROADS (AAR)

Specifications for Design, Fabrication and Construction of Freight Cars.
Manual of Standards and Recommended Practices:

Mechanical Division:

| | |
|---|---|
| Section A | - Specification for Materials. |
| Section D | - Trucks and Truck Details. |
| Section E | - Brakes, and Brake Equipment. |
| Section G | - Wheels. |
| Specification No. 2518 | - Installation, Freight Car Brake Equipment. |

Pamphlet No. 5039-4, Sup. 1 - Single Car Testing Device - Code of Test.
Supplement to the Manual of Standards and Recommended Practices.
United States Safety Appliances for All Classes of Cars and Locomotives.
Equipment Diagram, AAR Plate B for Unlimited Interchange Service.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren Street, Chicago, IL 60605.)

3. REQUIREMENTS

3.1 Description. The high-side gondola car shall have a 50-ton capacity. The cars shall be in accordance with AAR standards except as specified herein.



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3.2 First article.

3.2.1 First-produced cars. The contractor shall furnish one or more cars for examination and testing within the time frame specified (see 6.2), to prove that his production methods will produce cars that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.2).

3.3 Materials. Materials shall be as specified herein or equivalent ASTM specifications. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.3.1 Threaded parts. Threaded parts shall conform to FED-STD-H28.

3.3.2 Substitution of materials. Materials not conforming to the specifications referenced herein shall not be substituted: (1) Unless such substitutions are specifically approved by the contracting officer; (2) when it is clearly demonstrated that an improvement in operating characteristics, a saving in weight, conservation of critical or strategic materials, or a reduction in cost without sacrifice of reliability can be accomplished thereby; and (3) that such substitutions will not preclude the subsequent use of specified materials in effecting repairs or replacements. Requests for approval of such substitutions shall be accompanied by completely detailed supporting data.

3.4 Interchangeability. Jigs, templates, gages, and fixtures shall be used to insure interchangeability of like components and parts with respect to assembly, performance, and quality. All parts having the same part number shall be functionally and dimensionally interchangeable.

3.5 Tolerances and fits. Tolerances and fits shall conform to appropriate AAR Manuals applicable to railway cars unless otherwise specified herein.

3.6 Adjustments. All parts and components subject to adjustment or replacement by operating or maintenance personnel shall be accessible. Adjusting facilities where provided shall operate without interference and shall maintain such adjustments under all conditions of service.

3.7 Dimensions and weight. Dimensions and weight of the high-side gondola cars shall be as follows:

| | |
|---|---|
| Length over end sills | 42' - 0" (nominal) |
| Truck centers | 31' - 0" (nominal) |
| Truck wheel base | 5' - 6" |
| Height of sides, inside | 4' - 8" (nominal) |



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| | |
|---|---|
| Height of floor from top of rail | 3' - 8" (nominal) |
| Width over side sills | 9' - 6" (nominal) |
| Lightweight of cars | 42,000 lb. (nominal) |

3.7.1 Clearances. The cars shall not exceed the clearance limitations shown on AAR Plate B.

3.7.2 Curvature. The cars shall be capable of negotiating a 30-degree curve (193-foot radius) coupled.

3.8 Design and construction. Design and construction shall be in accordance with the AAR Specifications for Design, Fabrication and Construction of Freight Cars and AAR Supplement to the Manual of Standards and Recommended Practices, except as specified herein. When specified (see 6.2), structural analysis of the car in accordance with the specifications stated above shall be submitted by the car builder, as specified.

3.9 Underframe. The underframe shall be of welded construction.

3.9.1 Flooring. Flooring shall be nailable copper-bearing steel flooring with a minimum yield strength of 50,000 psi. Four drain holes shall be provided at the corners.

3.10 Trucks. Trucks shall be AAR-approved design, equipped with roller bearings, narrow pedestal type, 50-ton capacity, with built-in snubbing devices arranged to dampen vertical and lateral accelerations in accordance with AAR Manuals of Standards and Recommended Practices, Sections D and G, and AAR Wheel and Axle Manual, as applicable.

3.10.1 Side bearings. Side bearings shall be AAR approved double roller type.

3.10.2 Springs. Springs shall be 3-11/16 inch travel, AAR standard freightcar truck springs.

3.10.3 Wheels. AAR wheels shall be one-wear type. Wheels shall be mounted in accordance with the AAR Wheel and Axle Manual.

3.10.4 Axles. Axles shall be drilled and tapped on each end for roller-bearing retaining caps.

3.10.5 Axle bearings. Axle bearings shall be roller type, AAR approved design for standard narrow pedestal side frames.



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3.11 Brake system. The car shall be equipped with ABDW airbrake system. System shall conform to the AAR Manual of Standards and Recommended Practices, Section E. Schematic brake diagram indicating lever arrangement and braking forces should be furnished to the contracting officer prior to fabrication of badge plate. Installation shall be in accordance with Specification No. 2518, as adopted by the AAR.

3.11.1 Slack adjusters. Automatic slack adjusters, for the system, shall be AAR approved and shall have a range of travel which will permit brakeshoes to clear wheel tread surface with all shoes worn to limit or with all shoes new.

3.11.2 Handbrakes. The handbrake shall be an AAR approved, nonspin, vertical wheel, quick-release type.

3.11.3 Badge plate. A badge plate conforming to the AAR Manual of Standards and Recommended Practices shall be permanently attached to the underframe in an accessible location near the airbrake cylinder and on the same side of the car.

3.12 Couplers. Automatic couplers shall be AAR Standard Type E, rotary bottom operated with coupler operating levers at diagonal corners of car.

3.13 Superstructure. The car superstructure shall be the steel-sheathed outside post type. Side and end sheets shall be copper-bearing, steel plates. Side and end top chords shall extend in one length from end to end of car and across ends of car and have steel pressed corner cap at each corner. Corner posts shall be steel angles. End sills shall be fabricated from steel angles. Brake step shall be metal of AAR approved type. The car shall be equipped with lading strap anchors located between all side posts.

3.14 Safety appliances. Safety appliances shall conform to United States Safety Appliances for All Classes of Cars and Locomotives.

3.15 Routing card board. An AAR approved routing card board shall be applied to each side adjacent to the car number.

3.16 Placard board. A metal placard board may be applied to each side of the car.

3.17 Defect card holder. AAR type defect card holder shall be applied to the car.



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3.18 Identification plates. An identification plate conforming to MIL-P-3558, Type II, shall be permanently attached to the side sill of the car in a visible location.

3.19 Painting and marking. Painting shall be in accordance with MIL-P-3320, Type I, for Army, Navy and Air Force domestic car requirements. Marking shall be as specified in MIL-P-3320 and as required by the AAR for cars in interchange service. Registration numbers shall be as specified (see 6.2).

3.20 Lubrication. Roller bearings and truck center plates shall be lubricated in accordance with AAR Recommended Practices.

3.21 Workmanship. Workmanship for fabrication and construction shall be in accordance with AAR Specifications for Design, Fabrication and Construction of Freight Cars.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with applicable AAR specifications as modified herein or that the components have been previously approved by AAR.

4.1.2 Acceptability criteria. Cars which conform to all requirements in Sections 3 and 5 of this specification and pass all applicable examinations and tests in Section 4 of this specification will be considered acceptable by the Government.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First-produced car inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).



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4.3 First-produced car inspection. The first-produced car which shall have been produced from production tooling shall be examined and tested to determine conformance to the requirements of this specification.

4.3.1 Examination. The first-produced car shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first-produced car shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each car shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each car shall be tested as specified in 4.5.2.1 through 4.5.2.4. Failure of any test shall be cause for rejection.

4.5 Inspection procedures.

4.5.1 Examination. Each car shall be examined for the characteristics as specified herein as listed in Table I.

TABLE I - EXAMINATION SCHEDULE

| NUMBER | CHARACTERISTIC | REQUIRED PARAGRAPH |
|---|---|---|
| MAJOR | | |
| 101 | Material not as specified. | 3.3 |
| 102 | Threaded parts not as specified. | 3.3.1 |
| 103 | Parts and components not interchangeable, missing, or not as specified. | 3.4 |
| 104 | Tolerances and fits are not as specified. | 3.5 |
| 105 | Parts and components subject to replacement or adjustment not accessible. | 3.6 |
| 106 | Dimensions and weight not within specified limits. | 3.7 |
| 107 | Design and construction not as specified. | 3.8 |
| 108 | Brake equipment not as specified. | 3.11 |
| 109 | Safety appliances not as specified. | 3.14 |
| 110 | Routing card board not as specified. | 3.15 |
| 111 | Defect card holder not furnished or not applied as specified. | 3.17 |



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| NUMBER | CHARACTERISTIC | REQUIRED PARAGRAPH |
|---|---|---|
| 112 | Identification plate not furnished or not applied as specified. | 3.18 |
| 113 | Lubrication not as specified. | 3.20 |
| MINOR | | |
| 201 | Painting and marking not as specified. | 3.19 |
| 202 | Workmanship not as specified. | 3.21 |

4.5.2 Tests.

4.5.2.1 Test conditions. Tests shall be conducted in accordance with AAR requirements or as specified herein. Specific test procedures shall be submitted to the contracting officer for approval prior to testing. Prior to being tested, the car shall be lubricated with oil and grease as specified herein and put in operational condition.

4.5.2.2 Airbrake. The airbrake system shall be inspected and tested in accordance with AAR Instruction Pamphlet No. 5039-4, Sup. 1, Single Car Testing Device. Nonconformance to 3.11 or failure of this test shall be cause for rejection.

4.5.2.3 Handbrake. The handbrake shall be inspected and tested by applying and releasing the brakes to determine that the brake functions as specified (see 3.11.2) in accordance with AAR requirements. Nonconformance to 3.11.2 or AAR requirements shall constitute failure of this test.

4.5.2.4 Side bearings clearance. Side bearings clearance shall be checked to AAR standards. Nonconformance to 3.10.1 shall constitute failure of this test.

4.5.2.5 Slack adjuster. The slack adjuster shall be tested for range of travel. Nonconformance to 3.11.1 shall constitute failure of this test.

4.5.2.6 Curvature. The car shall be motivated through a 193-foot radius curve. At maximum swing, the brake rigging shall be inspected to determine that clearances have been provided and that the brake systems function correctly. Nonconformance to 3.7.2 shall constitute failure of this test. The test may be simulated by using the cross-track method.

4.6 Inspection of packaging.

4.6.1 Examination. Each car shall be examined for the following defect:

114. Car not prepared for delivery as specified in 5.1.



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5. PACKAGING

5.1 The cars shall be packaged in accordance with AAR requirements for interchange service.

6. NOTES

6.1 Intended use. The high-side gondola cars covered by this specification are intended to be used on domestic service for the transporting of commodities not requiring protection from the elements.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submittal of first-produced car (see 3.2.1).
(c) When AAR structural analysis is required (see 3.8).
(d) Registration numbers to be applied to car (see 3.19).

6.3 First produced car. Any changes or deviations of production cars from the approved first-produced car(s) during production will be subject to the approval of the contracting officer. Approval of the first-produced car(s) will not relieve contractor of his obligation to furnish cars conforming to this specification.

6.4 Recycled material. It is encouraged that recycled materials be used when practical as long as it meets the requirements of the specification (see 3.3).

Custodians:
Army - ME
Navy - YD
Air Force - 99

Preparing activity:
Army - ME

Project 2220-0135



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DOCUMENT IDENTIFIER AND TITLE MIL-R-3074D Railway Car, Gondola: High Side 56-1/2-Inch Gage, 50-Ton, 8-Wheel (Welded), Domestic Service

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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